Identification and classification of eyes movement using EEG signals

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***Abstract*— This work investigates the eye movement of 20 participants using the EEG signals measured using the headset wireless EMOTIV EPOC+ to collect EEG signals. The eye movement is detected analyzing the electroencephalographic (EEG) signals corresponding to four experiments: the participant follows with his eyes the movement of a circle on the screen. These movements are (i) from left to right (ii) from right to left (iii) from top to bottom and (iv) from bottom to top. The EEG signals obtained are classify using Support Vector Machine and Principal Component Analysis. The results obtained presents an accuracy between 81% and 91%. The above results show us that are possible to identify and classify the eye movements using EEG signals.**

Keywords—EEG, Brain Activity, BCI, Ocular Movement, Evoked Potential, SVM and PCA.

I. Introduction

Brain Computer Interfaces (BCI) represents a communication and control mechanism between the human brain and computers. Electrical activity of the brain, which is usually recorded using electroencephalogram (EEG). Is used to describe different cognitive and emotional states [7], and this activity patterns are involved in the execution and association of movements. Due with low cost and non-invasive character of EEG, which ones are the most popular techniques for monitoring brain activities in BCIs. There are two types of BCI: 1) Spontaneous based EEG, and 2) Evoked based EEG systems. The latest is generated as result as specific mental activity at specific areas as result of external stimulations. The aim of this type of BCI is generates distinct EEG signals watching some visual patterns with a Graphic User Interface (GUI), which are then translated into external actions. Previous works [17, 18], have reported that is possible to identify and classify with Neural Networks the different movements of the eyes using EEG signals. In this paper we identify a zone where is the neural activity and proposed an easy method with high accuracy between 81% to 91%.

The structure of this paper is organized as follows: Section 2 describes the data collection procedure including a description the EMOTIV EPOC+ headset based experiments. Section 3 describes the methods. Section 4 describes the preprocessing and feature extraction steps, and the use of information to investigate changes in EEG spectral activity in response of the visual patterns and we also identify and classify EEG signals. Presenting the experimental results; and finally, conclusions are provided in Section 5 interdependencies among the EEG signals from these regions; and (2) classify EEG activity using Support Vector Machine and Principal Component Analysis to classify the mental task. EEG has been used to describe different cognitive and emotional [9] in this paper we gone use EEG to describe the ocular movement.

II. Materials and Methods

A. Data Collection

An empirical study, using four test types was conducted consisting on the measurement of the EEG signal while the participant follows the motion of a circle on the screen (a) from left to right visual pattern, b) from right to left visual pattern c) from bottom to top visual pattern d) from top to bottom visual pattern).

B. Participants

The experiment was performed by 20 participants (10 males and 10 females) at the age of 18–41 years (average age 31). They all were healthy, physically active, with no neural disorders diagnosed. Some participants were right-handed and some left handed. The subjects had normal or corrected to normal vision and normal hearing. The methodology was fully explained to the participants who gave their consent to perform the experiment. On average, participants took 10 min to complete the experiment (i.e. explaining the instructions and then completing 5 trials of four set experiments). Moreover, the intended audience would be as general as possible, hoping to provide the level of heterogeneity necessary to validate the results.

The Visual patterns showed the type of motor imagery the subject is to execute for 10 s: (a) left- to- right visual pattern, (b) right to left visual pattern, (c) bottom to top visual pattern and (d) top to bottom visual pattern. Periods of around 2 s were introduced so that the subjects could take a short break. The data collection process employee one set of equipment the wireless headset with 14 channels. We are using EEG data as a source of information [1, 9]. Subjects were presented the visual patterns on a 23 screen at 20 cm with whole the electrodes at the right position.

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C. EMOTIV EPOC based EEG data collect

The EMOTIV is a high-resolution neuro signal acquisition and processing wireless headset that monitors 14 channels of EEG data and a gyroscope for two-dimensional control. The electrodes are located at the positions AF3, F7, F3, FC5, T7, P7, O1, O2, P8, T8, FC6, F4, F8, AF4 according to the International 10–20 system, forming 7 sets of symmetric channels as show Figure. 1. Two reference electrodes located just above the ears (one for the left hemisphere and the another one is for the right hemisphere of the head). The EMOTIV EPOC internally samples at a frequency of 2048 Hz, which then gets down-sampled to 128 Hz sampling frequency per channel, and sends via Bluetooth the data. Before to use the headset, all felt pads of the sensors must be moistened on top with a saline solution. The Emotive Software Development Kit (SDK) provides a packet count functionality to ensure no data is lost, a writable marker trace to ease single trial segmentation tasks, and real-time sensor contact display to ensure quality of measurements [8, 9]. The EMOTIV EPOC+ headset as a real-time brain EEG scanner was demonstrated in many recent publications, for that reason we use it in this paper.

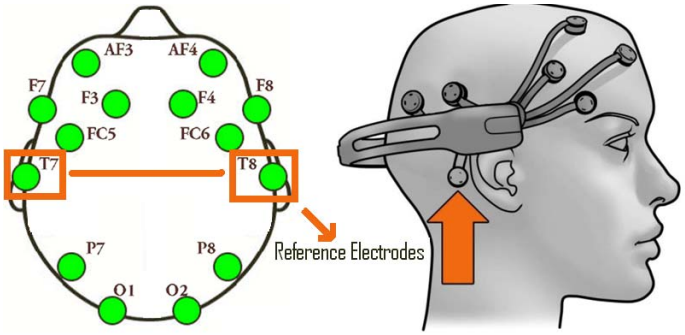


Figure 1 – EMOTIV EPOC+ electrode positioning on a subject showing left view.

D. Procedure

The study consists in four experiments. The first scenario was based on a ball appearing at the right border on the screen, after 2 seconds the ball starts to move to the left border in a linear movement where the movement describes the right visual pattern as show in Figure 2a. The second scenario was similar, the ball appears at the left border on the screen, after 2 seconds the ball starts to move to the right border, describing as the left visual pattern as show in Figure 2b. The third was modified changing some physical parameters, the ball starts at the low border on the screen, after 2 seconds the ball starts to move to the top border on the screen in a linear movement where the movement describes the up visual pattern as show in Figure 2c. The fourth scenario was similar, the ball appears at the upper border on the screen, after 2 seconds the ball starts to move to the low border, describing as the down visual pattern as show in Figure 2d. Velocity and time in Whole scenarios was measured as show in Table 2. The presented scenarios were one kind of checkerboards: black and white. The choice of used colors was determined by the high sensitivity of human eye on the contrast of those colors.

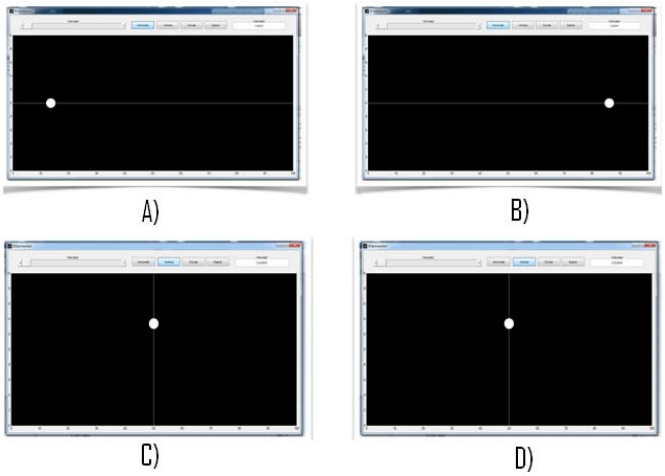


Figure 2- Illustration of the developed tasks a) Right visual pattern, b) Left visual pattern, c) Top visual pattern d) Bottom Visual pattern.

E. Data Description

We design a study of 4 visual patterns were used to evaluate performance and to identify difference on the ocular movement. This dataset contains 20 trials per person (100 trials for each task). Each trial lasts 10 s in a feedback session. In the beginning a ball displays there is 2 s of silence. At t = 2 s, beginning of trial was informed to the subject by a line "-" was displayed. Then at t = 3 s, the ball starts the left, right, top or bottom movement and the recording paradigm used shows in Fig. 3. The AF3, F7, F3, FC5, T7, P7, O1, O2, P8, T8, FC6. F4, F8, AF4 of EEG channels (accorded with the 10–20 International System) were used to classifiers should be continuous values (0 class 'left', 1 class 'right', 2 class 'up' and 3 class 'down') for each visual movement. The sign and magnitude of values specify class. A summary of information about each subject is presented in Table 1.

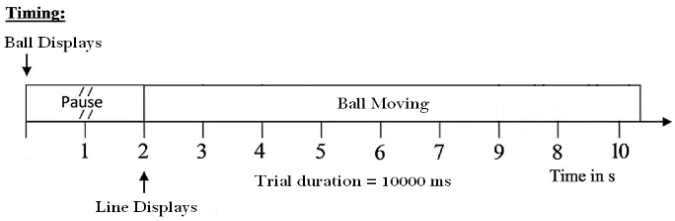


Figure 3 – Subject recording paradigm in a virtual experiment.

TABLE I. NUMBER OF TRIALS PER EXPERIMENT FOR EACH SUBJECT

Number of trials				
Subject	***Left Trials***	***Right Trials***	***Up Trials***	***Down Trials***
Male Subjects (H1, H2, H3, H4, H5, H6, H7, H8, H9, H10)	5 per person	5 per person	5 per person	5 per person
Female Subjects (M1, M2, M3, M4, M5, M6, M7, M8, M9, M10)	5 per person	5 per person	5 per person	5 per person
Total	100 per task	100 per task	100 per task	100 per task

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III. METHODS

A. Signal decomposition methods

In this study, two popular signal processing and decomposition techniques (Fourier Transform, Discrete Wavelet Transform), were used for the decomposition of Electroencephalography (EEG) Signals in Brain Computer Interface (BCI) system for a classification task. Those techniques are used to generate several sub-band signals by the fact that EEG frequency bands hold different energies during different imaging tasks [1,2,7]. In addition, statistical features calculated from these sub-band signals will express additional evidence about specific neural area activity. Signal processing techniques of EEG mainly belong to some of the following:

• Fourier transform (FT), which can be used to calculate power spectrum (or power spectral density), whose main disadvantages the lack of time domain information [1,7].

• Coefficients of discrete wavelet transform (DWT) at useful frequency bands. It decomposes the signal into the low-frequency and high-frequency components and then the coefficients at a certain band can be extracted [8].

Our paper represents a study providing a thorough comparison of signal classification methods in BCI signal analysis. The patterns of cortical activity were obtained in the five principal frequency bands, Delta (0–4 Hz), Theta (3–7 Hz), Alpha (8–12 Hz), Beta (13–30 Hz), and Gamma (30–40 Hz). There was a clear phase synchronization at the left frontal region indicating interhemispheric communications during the visual task for the 20 participants. Results also indicated that there was a clear and significant change among all the four experiments.

B. Classification

To validate propose tasks we implement two types of classifiers, Support Vector Machine (SVM) and Principal Component Analysis (PCA) are used in the experiments to compare with the signal decomposition methods.

B.1. Super Vector Machine classifier

SVM is a supervised machine learning method based on statistical learning theory which was introduced by Vapnik [15, 16]. SVM is a binary classifier which tries to find an optimal separating hyperplane between training data so that minimal error of classification and maximal margin from the support vectors (nearest training data to hyperplane) are obtained simultaneously. Training phase and finding separating hyperplane in classical SVM need a quadratic optimization problem to be solved which perform by a quadratic programming (QP) method [16,17]. In this study QP-SVM classifiers were used in experiments.

B.2. Principal Component Analysis

PCA is an algorithm capable of splitting linear relationship between the variables in the form of matrix of loadings. On the other hand, wavelets are used to split deterministic features from stochastic ones, resulting in approximate decorrelation of autocorrelation between the signals [7].

C. Method of performance evaluation

Some criteria are employed to analyze efficiency and performance of classification techniques. These criteria can be used to compare the effectiveness of parameter tuning in a different classifier with each other. The most widely used technique to help performance evaluating of classifier algorithms is Cross-Validation (CV). In this study, the K-fold CV and Holdout methods are used to evaluate average performance of classifiers.

D. EEG power spectrum analysis

We analyzed changes in spectral power and phase to characterize perturbations in oscillatory dynamics of ongoing EEG. Different amounts of frequency and amplitude seen at this analysis for the 4 tasks. The time spent by each participant was calculated from the data provided by the experiments and we calculated the total time across all participants by averaging the individuals' time as shown in Fig. 3.

IV. EXPERIMENTAL RESULTS

A. Feature Engineering

The data analysis procedure for measuring the differences brain activities between the four experiments at different locations described in the following sections. Using the mentioned instruments implied working with the EEG data of the four tasks. We focused on the ocular movement data, contained information related to the gaze position on the screen, pupil size and blink during the experiment. The data preprocessing consisted in two separate states 1) whole the data pre-processing was cleaned and filtered, and 2) the data pre-processing was cleaned and filtered and defined by channels. In both cases the data were filtered with a .5-6.3 Hz bandpass filter.

In a similar manner, several features were proposed. For the four tasks features were computed for each fixation, average voltage and range Frequency for all the participants as show Table 2.

TABLE II. AVERAGE AMPLITUDE AND RANGE FREQUENCY CORRESPONDING FOR THE FOUR EXPERIMENTS.

Amplitude and frequency without any filter		
Ocular Movements	***Average Voltage***	***Range Frequency***
Left	4.19 mV	0 – 29 Hz
Right	4.20 mV	0 -30 Hz
Down	3.59 μV	0 – 28 Hz
Up	3.54 μV	0 -31 Hz

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The EEG signals were analyzed using the Discrete Wavelet Transform (DWT) with two levels of decomposition (Alpha 4 - 8 Hz and Betha 16 – 32 Hz). The following features were then computed for each spectra and frequency band as show Figures 4 and 5.

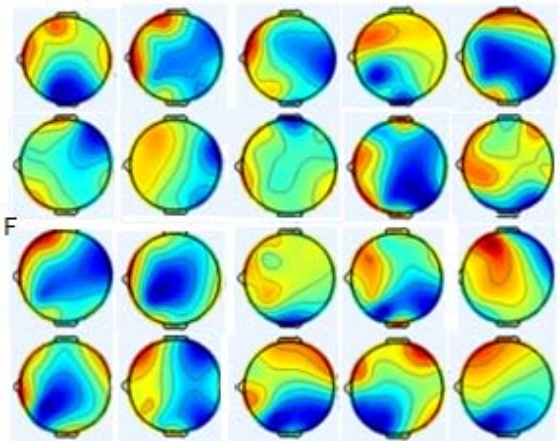


Figure 4 – Spectral Maps in a level of decomposition Alpha (4- 8 Hz).

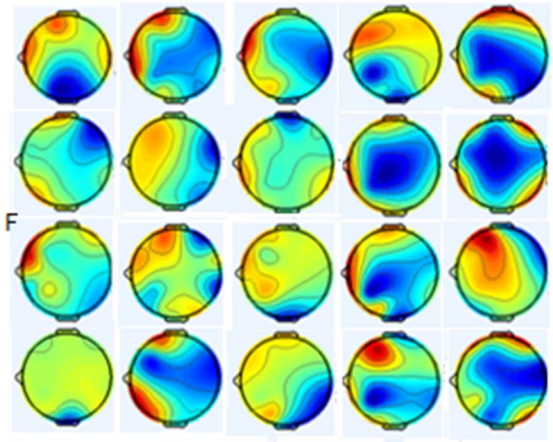


Figure 5 - Spectral Maps in a level of decomposition Betha (16- 32 Hz).

B. *Feature Classification*

Once the features were extracted, we classify with two classes because it is easier to visually the cases and which is the neural activity at each task. We used the data collected and filtered as inputs for the two methods on the feature selection, Support Vector Machine (SVM) by 10 folds (Figure 7 and 8) and Principal Component Analysis (PCA) with the fixed number of components 1, 10, 100, 300 and 500. We additionally compared both methods to the baseline. The results of this step are presented in Figure 9 and 10. We finally decided to use the level of decomposition Betha (16 – 32 Hz) because the neural activity when we classify gives us better accuracy and precision than the level of decomposition Betha (16 - 32 Hz) as show in Table 3.

TABLE III. LEVELS OF DECOMPOSITION WITH THE TWO METHODS OF CLASSIFICATION

Methods of Classification				
Ocular Movements	***SVM Alpha***	***SVM Betha***	***PCA-SVM Alpha***	***PCA-SVM Betha***
Left - Right	12%	63 %	50 %	92 %
Left - Bottom	15%	65%	56%	80%
Left - Top	12%	72%	56%	88%
Top - Bottom	21%	59%	59%	80%
Right - Bottom	13%	50%	44%	89%
Right - Top	15%	48%	60%	81%

For the PCA we found that we compare the ocular movements at the same axis as left-right or down-up is more difficult to identify but when we compare movements between different axis as left-bottom, left-top, right-bottom, and right-top is easier for PCA and SVM. We found another region where is possible to identify neural activity as show figure 6 and 7. We computed a grand average of data (Experimental data) for all subjects with the intention of finding patterns or differences between the four tasks. We found that for classification by SVM is the best option. This could be graphically seen in Fig. 6 and 7, where the dark green bar corresponds to a whole data and the light green, green and blue light and dark blue corresponds to SVM with PCA at 1, 10 100, 300 and 500 number of components. The X axis represents the fourteen channels and the Y axis represents the percentage of classification. Afterwards, we classified by PCA with SVM the six classification tasks (Left-Right, Left-Down, Left-Up, Up-Down, Right-Down and Right-Up). Table 8 and 9 shows the classification results, where it could be seen that the best performance in terms of Accuracy and Precision were obtained by the F8 and FC6 at whole experiments using PCA with SVM, because PCA keeps the linearity of the data and helps SVM to improve the classification method.

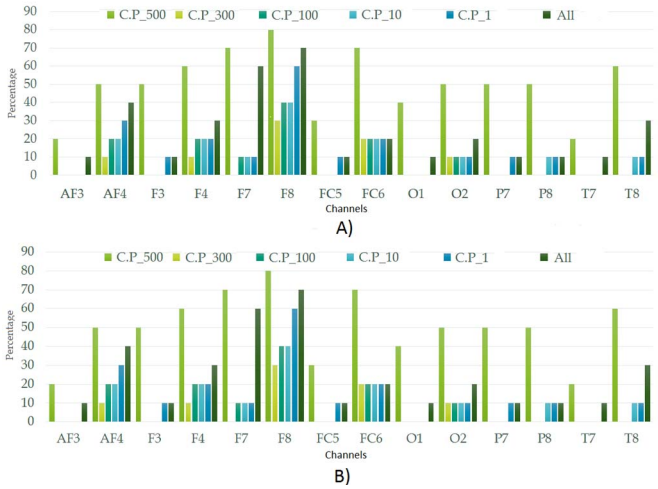


Figure 6 - Percentage values between all the classification methods per channel at cases: A) left-right (Horizontal) B) bottom-top (Vertical).

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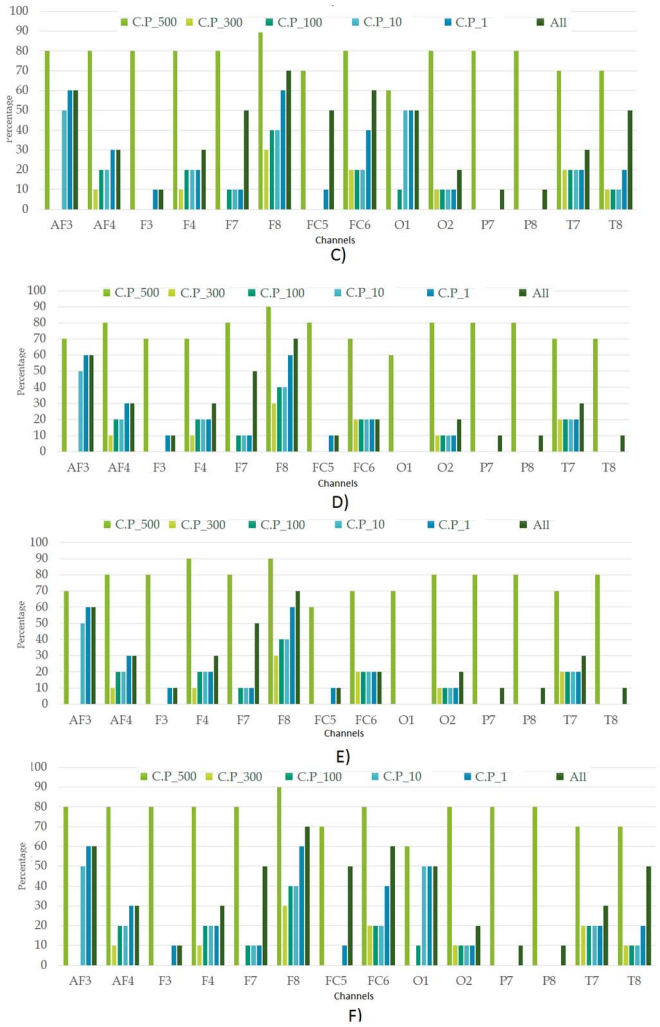


Figure 7 - Percentage values between all classification methods per channel at cases: C) left-bottom, D) left-top, E) right-bottom, and F) right-top.



Figure 8 - Percentage values between all the channels at cases: A) left-right (Horizontal) B) bottom-top (Vertical).

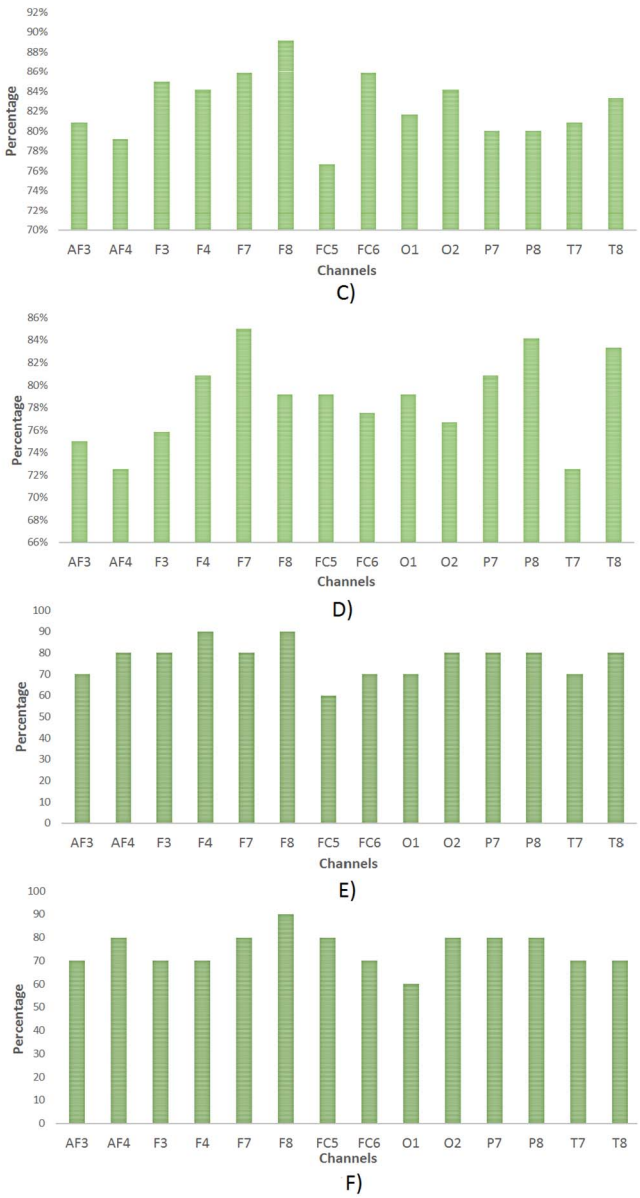


Figure 9 - Percentage values between all the channels at cases: C) left-bottom, D) left-top, E) right-bottom, and F) right-top.

V. Conclusions

In this work the behavior of the eye movement has been investigated considering the movement of the eyes following four visual patterns and measuring the electroencephalogram responses. For collecting the necessary data, we conducted an empirical study using four tasks (a) left to right visual pattern, b) right to left visual pattern c) top visual pattern d) bottom visual pattern). EEG data of 20 subjects were recorded while performing the tasks. Considering different aspects, we defined the data. The found as existence as statistical differentiation between the four tasks specifically that fixations corresponding to a close differentiation between the movement on the same axis. We used two levels of decomposition to corroborates our previous findings. In the EEG power spectral activities taking a

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place mainly in the frontal. We also proposed 6 classification models using 200 data per classification task (Left-Right, Left-Bottom, Left-Top, Top-Bottom, Right-Bottom and Right-Top) to classify the different methods using SVM and PCA. Although good results were obtained for Accuracy between 80% to 91% at the frontal (F8, FC6) regions. Additionally, the analysis indicated various factors as blinking or another visual distractor were important, affecting the experiments but these results show that is possible to identify the linear movements based EEG signals.

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REFERENCES

[1] Yeung, N., & Summerfield, C. (2012). Metacognition in human decision-making:Confidence and error monitoring. Philosophical Transactions of the Royal Society of London. Series B, Biological sciences, 367(1594), 1310–1321.

[2] Akira Hiraiwa, Katsunori Shimohora, Yukio Tokunaga "EEG Topography Recognition by Neural Networks", Engineering in medicine and Biology, September 1990.

[3] Rami N. Khushaba, Chelsea Wise, Sarath Kodagoda, Jordan Louviere, Barbara E. Kahn," Consumer Neuroscience Assesing the brain response to marketing stimuli using electroencephalogram (EEG) and eye tracking", 2012 International Journal Expert systems with Applications.

[4] Hamid Mirvaziri, Zabihollah Saberi Mobarakeh, "Improvement of EEG-based motor imaginary classification using ring topology-based particle swarm optimization", Biomedical Signal Processing and Control 32 (2017)69-75.

[5] Karina Maciejewska, Zofia Drzazga, Agniezka Trojankowska, "Visual Mismatch Negativity as a Non-Attentional Reaction to Change in Repetition Pattern", Springer International Publishing AG 2017.

[6] Brigida Monica Faria, Luis Paulo Reis, Nuno Lau," Cerebral Palsy EEG signals Classification: Facial Expressions and Thoughts for driving an Intelligent Chair" IEEE in press

[7] Jasmin Kevric, Abdulhamit Subasi, "Comparison of signal decomposition methods in classification of EEG signals for motor imagery BCI system", Biomedical Signal Processing and Control 31 (2017) 398-406.

[8] Rami N. Khushaba, Chelsea Wise, Sarath Kodagoda, Jordan Louviere, Barbara E. Kahn" Consumer Neuroscience Assesing the brain response to marketing stimuli using electroencephalogram (EEG) and eye tracking".

[9] Bobrov, P., Frolov, A., Cantor, C., Fedulova, I., Bakhnyan, M., & Zhavoronkov, A. (2011). Brain-computer interface based on generation of visual images. PLoSONE, 6(6), e20674, pp. 1–12.

[10] J.R. Wolpaw, N. Birbaumer, D.J. McFarland, G. Pfurtscheller, T.M. Vaughan, Brain–computer interfaces for communication and control, Clin. Neurophysiol. 113 (2002) 767–791.

[11] F. Lotte, M. Congedo, A. Lécuyer, F. Lamarche, B. Arnaldi, A review of classification algorithms for EEG-based brain?computer interfaces, J. Neural Eng. 4 (2007) 1–13.

[12] R. Fazel-Rezai, Recent Advances in Brain-Computer Interface Systems, InTech, Croatia, 2011.

[13] Q. Xu, H. Zhou, Y. Wang, J. Huang, Fuzzy support vector machine for classification of EEG signals using wavelet-based features, Med. Eng. Phys. 31 (2009) 858–865.

[14] Kimura, M., Kondo, H., Ohira, H., Schroger, E.: Unintentional temporal context based prediction of emotional faces: an electrophysiological study. Cereb. Cortex 22, 1774–1785 (2012)

[15] Huettel, S.A., Mack, P.B., McCarthy, G.: Perceiving patterns in random series: dynamic processing of sequence in prefrontal cortex. Nat. Neurosci. 5, 485–490 (2002)

[16] G. Rodríguez-Bermúdez, P.J. García-Laencina, Automatic and adaptiveclassification of electroencephalographic signals for brain computerinterfaces, J. Med. Stems 36 (2012) S51–S63.

[17] Sherif M. Abdelfattah, Kathryn E. Merrick, Hussein A. Abbass, "Eye movements as information markers in EEG data", Computational Intelligence (SSCI), 2016 IEEE Symposium Series.

[18] Aasim Raheel, Syed M. Anwar, Muhammad Majid, "Real time text speller based on eye movement classification using wearable EEG sensors", SAI Computing Conference (SAI), 2016.

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